Estimand Alignment and Outcome-Specific Discrepancies in a Four-Arm Within-Study Comparison: A Concordant-Population Benchmark and Decomposition of Observational–Experimental Differences

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Abstract

Within-study comparisons require randomized and observational estimates to target the same causal parameter. We derive a randomized benchmark for the concordant population in a four-arm preference design and show that it equals the full-population average treatment effect when the common assignment probability is one half or preference-group effects are equal. In the WSC sample (N = 2,200), the prespecified mathematics post-test comparison pairs the randomized effect among mathematics-preferring participants with a cross-fitted observational average treatment effect on the treated in the concordant subset. Estimates are 0.795 and 0.794 (difference −0.00045; 95% paired-bootstrap interval [−0.949, 0.664]). A finite-sample decomposition of the full-sample discrepancy (0.315 points) assigns 0.288 to the observational ATE–concordant benchmark gap and 0.027 to the difference between randomized estimators; under the published one-half assignment design, the population composition term is zero. In a post hoc vocabulary-outcome analysis targeting the same mathematics-preferring population, the observational ATU exceeds its randomized benchmark by 0.913 (nominal 95% bootstrap interval [0.115, 1.909]). This contrast suggests that observational agreement is outcome-dependent and warrants exploratory interpretation.

Keywords: within-study comparison; estimand alignment; concordant population; outcome-specific discrepancy; average treatment effect; cross-fitting.

1 Introduction

Within-study comparisons place a randomized estimate and a nonrandomized estimate in one empirical setting, so that differences of site, intervention, or outcome instrument are less available as explanations for a discrepancy (Keller et al., 2025). The comparison still has to say which causal parameter each estimate targets. A randomized contrast among people who preferred treatment, an average effect in the subset whose assignment matched their preference, and an average effect in the full sample can all be well defined and still be different parameters. Calling every gap between an observational estimate and a randomized estimate "bias" collapses three separate ideas: whether two estimands are the same, whether an observational functional identifies its target, and how precisely either quantity is estimated.

The public four-arm study of Keller et al. (2025) makes this distinction concrete. Participants stated a preference for a mathematics or vocabulary session before being randomly assigned to one of the two sessions. The published design description specifies assignment probability one half, independent of preference and potential outcomes. The concordant population consists of participants whose assignment matches their preference. Within this population, the randomized treatment

Table 1: Realized four-arm cells. S is preference for mathematics and D is assignment to mathematics.

Cell	Definition	Role in $C = \{D = S\}$	n
G1	$S = 1$, $D = 1$	Concordant, assigned mathematics	288
G2	$S = 1$, $D = 0$	Discordant	319
G3	$S = 0$, $D = 1$	Discordant	776
G4	$S = 0$, $D = 0$	Concordant, assigned vocabulary	817

groups correspond exactly to the two preference groups, providing randomized benchmarks for both preference-specific effects and their concordant-population average. An observational analysis of the same participants, however, uses preference as the treatment and therefore requires assumptions about self-selection.

This paper develops a concordant-population benchmark, relates it to the full-population average treatment effect, and examines how observational estimates compare with randomized benchmarks when the target population and estimand are explicitly aligned. The empirical analysis uses the public WSC sample of 2,200 participants. Its prespecified primary comparison pairs the randomized difference in means among mathematics-preferring participants with a cross-fitted augmented inverse-probability-weighted estimate of the average treatment effect on the treated in the concordant subset (Bang and Robins, 2005; Chernozhukov et al., 2018). A finite-sample decomposition then separates the full-sample observational-minus-randomized difference into the observational ATE–concordant benchmark gap, a sample composition term, and the difference between two randomized estimators.

A post hoc analysis applies the same concordant-sample adjustment strategy to the vocabulary post-test, reversing the treatment and preference indicators so that the target population remains mathematics-preferring participants. The mathematics and vocabulary results show different patterns: the prespecified mathematics point estimates are nearly identical, whereas the vocabulary ATU estimate is higher than its randomized benchmark. The paper therefore contributes a design-based estimand result and an outcome-specific empirical comparison, while distinguishing randomized identification of the benchmarks from observational identification of the corresponding effects.

2 Study design and data

Let $S \in \{0, 1\}$ denote preference for the mathematics session, and let $D \in \{0, 1\}$ denote randomized assignment to that session. The concordant population is $C = \{D = S\}$. The four realized cells are G1, $(S, D) = (1, 1)$; G2, $(1, 0)$; G3, $(0, 1)$; and G4, $(0, 0)$. Table 1 gives the counts in the public file: 288, 319, 776, and 817. Mathematics assignment was realized for $288 + 776 = 1{,}064$ participants and vocabulary assignment for $319 + 817 = 1{,}136$. The analysis sample is the released WSCdata file of $N = 2{,}200$ rows described by Keller et al. (2025). The prespecified primary outcome is the mathematics post-test, `mathPost`; a post hoc exploratory analysis also examines the vocabulary post-test, `vocabPost`.

Under the published description of the WSC randomization design, preference was elicited before assignment, and assignment was then made regardless of preference (Keller et al., 2025). The description specifies complete random assignment with a common probability of one half and no preference-based stratification or blocking. In the released sample, 1,064 of 2,200 participants were assigned to mathematics; a two-sided exact binomial test against probability one half gives $p = 0.130$. The association between assignment and preference is also small (Yates-corrected Pearson $\chi^2 = 0.234$, $p = 0.629$). These descriptive checks are compatible with the published design

description. The assignment program was not inspected, so the design assumption is taken from the published account.

The primary randomized contrast uses G1 and G2, the two cells with $S = 1$. The primary observational contrast uses the concordant subset G1∪G4, which contains 1,105 participants. The full-sample randomized difference in means uses all four cells. The preference-group effect for $S = 0$ uses G3 and G4. These are different samples, and the parameters attached to them are defined in the next section before any estimator is introduced.

3 Causal estimands

Let $Y(1)$ and $Y(0)$ denote the potential mathematics post-test outcomes under assignment to mathematics and to vocabulary. The preference-group effects, the full-population average treatment effect, and the concordant-population effect are

$$\begin{aligned} \tau_s &= E[Y(1) - Y(0) \mid S = s], \qquad s \in \{0, 1\}, \\ \tau_{\text{full}} &= E[Y(1) - Y(0)], \\ \tau_C &= E[Y(1) - Y(0) \mid C]. \end{aligned}$$

Write $p = P(D = 1)$, $\pi = P(S = 1)$, and $\pi_c = P(S = 1 \mid C)$. The average treatment effect on the treated in the usual selection sense is $E[Y(1) - Y(0) \mid S = 1] = \tau_1$, because the people who prefer mathematics are the people who would be treated under self-selection. Likewise τ_0 is the average treatment effect on the untreated. Inside C, those two parameters are also the effects in the assigned-treated and assigned-untreated concordant strata.

The design assumptions used for the population identities are the following.

1. **Random assignment.** $D \perp \big(S, Y(1), Y(0)\big)$, with a common assignment probability $0 < p < 1$ in both preference groups.

2. **Both preferences occur.** $0 < \pi < 1$.

3. **Consistency and no interference.** The observed outcome equals the potential outcome under the realized assignment; there is one treatment version; and one participant's assignment does not change another's potential outcomes. The further statement that randomized assignment and self-selection implement the same treatment version is part of this assumption. It is not implied by independence of D.

Assumption 1 is the content of the published randomization description. Assumption 3 is stated separately by Keller et al. (2025) and is retained here as an assumption. The identities in Proposition 1 use Assumptions 1 and 2. They are statements about potential-outcome estimands. Consistency is what connects those estimands to contrasts of observed outcomes.

Proposition 1. *Under Assumptions 1 and 2,*

$$\pi_c = \frac{\pi p}{\pi p + (1 - \pi)(1 - p)}, \tag{1}$$

$$E[Y(1) - Y(0) \mid D = 1,\, C] = \tau_1, \tag{2}$$

$$E[Y(1) - Y(0) \mid D = 0,\, C] = \tau_0, \tag{3}$$

$$\tau_C = \pi_c \tau_1 + (1 - \pi_c)\tau_0, \tag{4}$$

$$\tau_C - \tau_{\text{full}} = (\pi_c - \pi)(\tau_1 - \tau_0). \tag{5}$$

Consequently,

$$\tau_C = \tau_{\text{full}} \quad \Longleftrightarrow \quad p = \tfrac{1}{2} \quad \textit{or} \quad \tau_1 = \tau_0. \tag{6}$$

Proof. Independence of D and S gives $P(S=1, D=1) = \pi p$ and $P(S=0, D=0) = (1-\pi)(1-p)$. The event C is the disjoint union of these two cells, so (1) is the definition of π_c. The event $\{D=1\} \cap C$ equals $\{D=1, S=1\}$. Independence of D and $(Y(1), Y(0))$ given $S=1$ therefore yields (2), and the symmetric argument yields (3). Conditioning further on S inside C,

$$\tau_C = E[Y(1) - Y(0) \mid S=1,\, C]\, \pi_c + E[Y(1) - Y(0) \mid S=0,\, C]\, (1-\pi_c).$$

The two conditional expectations are τ_1 and τ_0, which is (4). The full-population effect has the same two building blocks with weights $(\pi, 1-\pi)$,

$$\tau_{\text{full}} = \pi\tau_1 + (1-\pi)\tau_0,$$

and this decomposition uses only the law of total expectation. Subtracting it from (4) gives (5). Finally,

$$\pi_c - \pi = \frac{\pi(1-\pi)(2p-1)}{\pi p + (1-\pi)(1-p)}.$$

The denominator is positive under Assumptions 1 and 2, and $\pi(1-\pi) > 0$, so $\pi_c = \pi$ if and only if $p = 1/2$. Combined with (5), this is (6). □

The concordant benchmark can be assembled from the two randomized preference-group effects without first estimating p. Under the published one-half assignment design, $\pi_c = \pi$ and hence $\tau_C = \tau_{\text{full}}$ at the population level. For other common assignment probabilities, τ_C remains a randomized benchmark for the concordant population, while equality with the full-population effect additionally requires homogeneous preference-group effects.

The observational comparison in C has a separate identification requirement. Because $D = S$ within the concordant subset, estimating the preference-group effects from G1∪G4 requires conditional exchangeability of preference given the 25 covariates, positivity, and the consistency, no-interference, and common-treatment-version assumptions in Assumption 3.

4 Estimation and inference

The randomized building blocks are unadjusted differences in means. For a binary assignment indicator and an observed outcome, the estimator is the treated mean minus the control mean, with the usual two-sample standard error formed from the two sample variances. Applied to G1 versus G2, this estimates τ_1. Applied to G3 versus G4, it estimates τ_0. Applied to all 2,200 rows, it estimates the full-sample average effect of assignment. The sample assignment proportion is left as realized; the full-sample estimator is not reweighted to one half. Write $\widehat{\tau}_1$, $\widehat{\tau}_0$, and $\widehat{\tau}^{\text{RCT}}_{\text{full,DM}}$ for these three statistics, and write $\widehat{\pi} = 607/2200$ and $\widehat{\pi}_c = 288/1105$ for the realized preference weight and the realized concordant preference weight. The plug-in benchmarks are

$$\widehat{\tau}_C = \widehat{\pi}_c\, \widehat{\tau}_1 + (1 - \widehat{\pi}_c)\, \widehat{\tau}_0,$$
$$\widehat{\tau}_{\text{ps}} = \widehat{\pi}\, \widehat{\tau}_1 + (1 - \widehat{\pi})\, \widehat{\tau}_0.$$

The first is the finite-sample concordant benchmark. The second is the preference-post-stratified full-sample benchmark. Their difference equals $(\widehat{\pi}_c - \widehat{\pi})(\widehat{\tau}_1 - \widehat{\tau}_0)$.

The observational estimator in the concordant subset is a five-fold cross-fit augmented inverse-probability-weighted functional (Bang and Robins, 2005; Chernozhukov et al., 2018). The propensity score is the conditional probability of mathematics assignment given covariates (Rosenbaum and Rubin, 1983). In C that event coincides with preference for mathematics. The nuisance functions are an unpenalized logistic regression for the propensity score and linear regressions for the outcome in each arm. Out-of-fold predictions are clipped to $[10^{-6}, 1 - 10^{-6}]$, and no row is deleted because of the estimated propensity score. Preference is not used as a covariate. The adjustment set has 25 baseline variables: demographics and schooling, the two pre-tests, mathematics motivation and anxiety measures, and personality and related psychological scales. The primary functional is the average treatment effect on the treated, evaluated on G1∪G4, so that its randomized counterpart is τ_1. The cross-fitting seed for that prespecified fit is fixed. The analytic standard error reported with the primary observational estimate is the standard deviation of the cross-fit scores divided by the square root of the concordant sample size. That standard error does not repeat the nuisance-function fit inside a resampling distribution.

The prespecified uncertainty statement for the primary contrast is a paired bootstrap. Each of 2,000 replicates redraws G1, G2, and G4 with replacement at their observed sizes, holds out G3, uses the same G1 draw in the randomized and observational estimators, refits the observational nuisance functions, and records the observational estimate minus $\widehat{\tau}_1$. All 2,000 replicates produced an estimate. The reported interval is the 2.5th and 97.5th percentiles of those 2,000 differences. This interval is the primary inferential result. A later implementation used a different random-number stream, redrew all four cells, and produced a different percentile interval for the same contrast. That calculation is reported in the supplement as an implementation check. It does not replace the prespecified interval.

The post hoc vocabulary analysis defines $D^V = 1 - D$ (assignment to vocabulary), $S^V = 1 - S$ (preference for vocabulary), and $Y^V = \texttt{vocabPost}$. The concordant subset is unchanged because $D^V = S^V$ if and only if $D = S$. The same 25 covariates, five-fold cross-fitting specification, and seed 20261026 are used. Under the reversed coding, the QED ATU targets participants who prefer mathematics (G1), and its randomized counterpart is the vocabulary-minus-mathematics effect among those same participants. With the same symmetric logistic specification and fold assignment, reversing the treatment label gives $\widehat{e}_V(X) = 1 - \widehat{e}_M(X)$. Thus, for G4, the vocabulary ATU weight $(1 - \widehat{e}_V)/\widehat{e}_V$ equals the mathematics ATT weight $\widehat{e}_M/(1 - \widehat{e}_M)$. The supplied weight summary reports an effective sample size of 145.9 and a maximum weight of 7.70, matching the mathematics ATT diagnostics. The vocabulary contrasts use a four-cell bootstrap with 2,000 resamples, because the randomized preference-group benchmarks require both concordant and discordant cells. This outcome analysis was conducted after the primary mathematics results and is exploratory.

The decomposition in Section 5 uses one joint redraw of all four cells, again at the observed cell sizes, with every estimator recomputed on the same redraw and with failed replicates retained rather than replaced. In that bootstrap the cell sizes, and therefore $\widehat{\pi}$ and $\widehat{\pi}_c$, stay fixed. The procedure conditions on the observed design margins. It is not a rerandomization test, and it does not estimate the sampling distribution that would arise if the preference counts were redrawn from a superpopulation. Component intervals from this bootstrap are secondary. They are attached to the decomposition, not to the primary contrast.

An algebraic identity organizes the full-sample discrepancy. Let $\widehat{\tau}^{\mathrm{QED}}_{\mathrm{ATE}}$ denote the cross-fit augmented inverse-probability-weighted average treatment effect on G1∪G4. Then

$$\widehat{\tau}^{\mathrm{QED}}_{\mathrm{ATE}} - \widehat{\tau}^{\mathrm{RCT}}_{\mathrm{full,DM}} = (\widehat{\tau}^{\mathrm{QED}}_{\mathrm{ATE}} - \widehat{\tau}_C) + (\widehat{\tau}_C - \widehat{\tau}_{\mathrm{ps}}) + (\widehat{\tau}_{\mathrm{ps}} - \widehat{\tau}^{\mathrm{RCT}}_{\mathrm{full,DM}}). \tag{7}$$

The first term compares the observational average treatment effect with the concordant randomized benchmark. It mixes sampling error, outcome-model and propensity-model error, and whatever

Table 2: Prespecified primary comparison for `mathPost`. For the first two rows, the standard error is analytic and the interval is the corresponding normal interval. For the contrast, the standard error is the standard deviation of the prespecified 2,000 bootstrap replicates and the interval is their percentile interval.

Estimand	Estimate	SE	95% interval	Sample
Randomized τ_1 (G1 vs. G2)	0.795	0.278	[0.250, 1.339]	288 / 319
Observational ATT (G1∪G4)	0.794	0.224	[0.356, 1.233]	288 / 817
Observational minus randomized	−0.00045	0.422	[−0.949, 0.664]	paired bootstrap

selection bias remains after adjustment. The second term is estimand composition: the concordant benchmark uses $\widehat{\pi}_c$, while the post-stratified benchmark uses $\widehat{\pi}$. The third term compares two randomized estimators of full-population quantities, the post-stratified combination of preference-group differences in means and the unadjusted full-sample difference in means. None of the three terms is, by itself, a confirmed causal bias.

5 Results

5.1 Prespecified primary comparison

Table 2 reports the primary pair. Among the 607 mathematics-preferring participants, the randomized difference in means is 0.795 points, with analytic standard error 0.278 and analytic normal interval [0.250, 1.339]. In the concordant subset, the cross-fit observational estimator of the average treatment effect on the treated is 0.794, with score standard error 0.224 and analytic interval [0.356, 1.233]. The observational estimate minus the randomized estimate is −0.00045. The prespecified percentile interval is [−0.949, 0.664], and the bootstrap standard deviation of the contrast is 0.422.

The two point estimates agree through the third decimal place. The paired-bootstrap interval is substantially wider than their point difference, spanning −0.949 to 0.664. The cross-fit score standard error (0.224) and paired-bootstrap standard deviation (0.422) summarize uncertainty through different procedures: the latter resamples the cells and refits the nuisance functions. Of the 1,105 concordant propensity scores, 396 (35.8 percent) lie outside [0.1, 0.9]. The ATT control weights have an effective sample size of 145.9 among 817 controls (17.9 percent); no observation was trimmed in the primary analysis.

A second implementation reproduced the two primary point estimates to within 1.3×10^{-15}. Its four-cell bootstrap interval was approximately [−0.925, 0.661], with standard deviation 0.470. The prespecified interval in Table 2 remains the primary uncertainty statement.

5.2 Concordant average treatment effect

The concordant randomized benchmark $\widehat{\tau}_C$ equals 0.787, while the cross-fit observational ATE on the same 1,105 participants equals 1.075. Their difference is 0.288, with a four-cell bootstrap interval of [−0.634, 1.073]. This secondary comparison aligns the observational ATE with the randomized effect in the concordant population.

The observational ATE exceeds the unadjusted full-sample randomized difference (0.760) by 0.315. Under the published one-half assignment design, the population parameters τ_C and τ_{full}

Table 3: Finite-sample decomposition of the full-sample discrepancy. The three components sum to the total at the displayed precision. The percentile interval is from the four-cell bootstrap and conditions on the observed cell sizes.

Component	Value
Observational ATE, $\widehat{\tau}_{\text{ATE}}^{\text{QED}}$	1.0753974159
Full-sample randomized difference in means	0.7603449645
Total discrepancy	0.3150524514
Observational ATE minus concordant benchmark, $\widehat{\tau}_{\text{ATE}}^{\text{QED}} - \widehat{\tau}_C$	0.2882422278
Estimand composition, $\widehat{\tau}_C - \widehat{\tau}_{\text{ps}}$	−0.0001555014
Post-stratified minus unadjusted randomized estimator	0.0269657250
Concordant benchmark, $\widehat{\tau}_C$	0.7871551881

coincide. The finite-sample difference is therefore examined through the estimator decomposition below, rather than attributed to a population-level estimand mismatch.

5.3 Finite-sample decomposition

Table 3 enters the three terms of (7) at the precision needed to check the sum. The observational average treatment effect is 1.0753974159 and the full-sample randomized difference in means is 0.7603449645. Their difference, 0.3150524514, equals

$$0.2882422278 + (-0.0001555014) + 0.0269657250.$$

The unrounded values satisfy the same identity with a numerical remainder of zero. Across the 2,000 joint bootstrap replicates, the largest absolute departure from the row-wise identity is 4.4×10^{-16}.

Under the published $p = 1/2$ design, the population composition term is zero. Its finite-sample plug-in estimate is −0.00016, reflecting the small difference between the realized preference weights $\widehat{\pi}_c$ and $\widehat{\pi}$. The four-cell bootstrap fixes that weight difference; the corresponding conditional interval for $\widehat{\tau}_1 - \widehat{\tau}_0$ is approximately [−0.641, 0.622], with point estimate 0.0102.

The third term, 0.027, is the finite-sample difference between the post-stratified randomized estimate ($\widehat{\tau}_{\text{ps}} = 0.787$) and the unadjusted full-sample difference in means (0.760). Its conditional percentile interval is [0.024, 0.030].

The leading term, 0.288 with interval [−0.634, 1.073], is the gap between the observational ATE and the concordant randomized benchmark. It combines sampling variation, nuisance-model estimation error, and any residual selection bias.

6 Exploratory analyses

The analyses in this section were not part of the prespecified primary comparison. They use the same public sample after the primary estimates were known. Numerical detail, including the implementation-specific primary bootstrap, is in the Supplementary Material. Supplementary Figures S1–S4 display the joint-bootstrap margins, the cross-fitting seeds, the nested covariate sets, and the sensitivity bounds.

6.1 Average treatment effect on the untreated

The randomized benchmark $\widehat{\tau}_0$ is 0.785, and the cross-fit observational estimate of the corresponding untreated effect is 1.175. Their difference is 0.390, with four-cell percentile interval [−0.755, 1.479]. The point estimates are asymmetric relative to the primary ATT comparison. The untreated weights are more concentrated: the effective sample size is 61.9 of 288 treated-concordant participants (21.5 percent), the maximum weight is 52.57, and the ten largest weights account for 31.3 percent of total weight.

6.2 Vocabulary post-test replication

The vocabulary analysis reverses the treatment and preference labels and replaces `mathPost` with `vocabPost`. Its concordant sample remains G1∪G4. The comparison for mathematics-preferring participants is the randomized vocabulary-minus-mathematics effect in G1 versus the QED ATU under the reversed coding. Table 4 reports all three concordant estimand comparisons.

Table 4: Post hoc vocabulary post-test comparisons. Differences are QED minus randomized; intervals are nominal 95% percentile intervals from the four-cell bootstrap.

Target population	Randomized	QED	Difference	Bootstrap SE	95% interval
Vocabulary-preferring (ATT)	2.185	2.577	0.392	1.070	[−2.066, 2.197]
Mathematics-preferring (ATU)	2.039	2.952	0.913	0.463	[0.115, 1.909]
Concordant population (ATE)	2.147	2.674	0.528	0.799	[−1.254, 1.841]

For the mathematics-preferring target population, the QED ATU exceeds the randomized benchmark by 0.913 vocabulary-test points, approximately 45 percent of the randomized benchmark point estimate. Its nominal 95% interval excludes zero. The ATT and concordant ATE contrasts are less precise and their intervals include zero. The ATU weighting uses the same inverse-odds weighting direction as the mathematics-outcome ATT comparison; the supplied diagnostics report an effective sample size of 145.9 and a maximum weight of 7.70. Thus the two outcome analyses compare the same preference group under the same covariate adjustment and corresponding weighting construction.

The contrast across outcomes is consistent with outcome-specific differences in residual selection or outcome-model error. The analysis was conducted post hoc, and the six contrasts across two outcomes and three estimands were not adjusted for multiplicity; the interval should therefore be read as an exploratory signal rather than a confirmatory test. Supplementary Table S3 reports the corresponding estimates and bootstrap summaries.

6.3 Cross-fitting partition

Repeating the observational fit at 200 fold seeds on the same participants gives a median treated-effect estimate of 0.734. The absolute contrast with $\widehat{\tau}_1$ is below 0.01 for 11.5 percent of these partitions, and the prespecified seed's estimate is at the 77th percentile. The spread describes sensitivity to fold assignment within this sample.

6.4 An alternative randomized estimator

A linear adjustment that interacts assignment with centered covariates, using an HC2 variance estimator, is a second randomized estimator of the same preference-group effects (Lin, 2013). It

produces $\widehat{\tau}_1 = 0.695$ and $\widehat{\tau}_0 = 0.810$, compared with 0.795 and 0.785 for the unadjusted differences in means. Both estimators are valid under the randomization; their finite-sample differences illustrate sensitivity to the choice of randomized estimator.

6.5 Covariate sets

Nested fits add demographics, pre-tests, mathematics-motivation measures, and psychological scales; each nonempty estimate is summarized by its median over 50 fold seeds. With no covariates, all three observational functionals equal the concordant difference in means, 3.451. With the full set, median discrepancies from the randomized ATT, ATU, and ATE benchmarks are -0.056, 0.425, and 0.304. Omitting the mathematics pre-test raises the median treated estimate by 0.669, while omitting mathematics liking raises it by 0.249. These patterns are consistent with the importance of pretreatment measures closely related to selection and outcome (Steiner et al., 2010); the leave-one-out shifts are model-sensitivity diagnostics rather than an additive decomposition of selection bias. The full-set median treated estimate, 0.738, differs from the prespecified estimate, 0.794, because the latter uses one fixed fold partition.

6.6 Marginal sensitivity bounds

A marginal sensitivity model tilts the odds of treatment, relative to the fitted propensity-score odds, by at most a factor Λ (Tan, 2006; Zhao et al., 2019). The reported worst-case bounds apply to a Hájek inverse-probability-weighted functional under that restriction; they are neither the sharp IPW bounds of Dorn and Guo (2023) nor bounds for the augmented estimator. At $\Lambda = 1$, the IPW treated and untreated estimates are 0.686 and 0.960. The augmented untreated estimate is 1.175, about 0.215 above the IPW value, reflecting the contribution of the outcome regression.

On a grid with steps of 0.01, the first Λ at which the IPW lower bound for the untreated effect falls to or below the randomized point estimate $\widehat{\tau}_0$ is 1.06. At $\Lambda = 1.10$, both randomized point estimates lie within the point bounds. Percentile intervals for the bound endpoints summarize their sampling variation and are not confidence intervals for a causal parameter.

6.7 Overlap and trimming

Trimming on the propensity score changes the population whose effect is being estimated (Crump et al., 2009). Here the propensity score used to define regions was fit once on the concordant covariate distribution, without the outcome, and then applied to all 2,200 rows. The bootstrap for a trimmed region does not refit that trimming model. Under random assignment, the randomized and observational estimators on a region defined by covariates target a common restricted population, and that population is not the original one. In the region with scores in [0.1, 0.9], the randomized $\widehat{\tau}_0$ falls from 0.785 to 0.430, and the untreated observational contrast moves from 0.390 to 0.716. The Crump threshold selected on the concordant sample is $\alpha = 0.084$, leaving 1,561 participants. These shifts are exploratory descriptions of different populations.

The untrimmed row of that trimming exercise uses a separate 1,000-replicate bootstrap. Its interval for the treated contrast is approximately $[-1.004, 0.682]$. That interval is not the primary interval in Table 2, and it is not the 2,000-replicate implementation-check interval $[-0.925, 0.661]$.

7 Discussion

The prespecified mathematics comparison produces nearly identical randomized and observational point estimates: 0.795 and 0.794. The paired-bootstrap interval for their difference, however, spans $[-0.949, 0.664]$. The result is a close point-estimate comparison with substantial uncertainty, not a precise equivalence finding.

The design result is broader than the primary ATT comparison. The concordant-population benchmark can be constructed from the two randomized preference-group effects for any common assignment probability in $(0, 1)$. Under the published one-half assignment design, it equals the full-population average treatment effect. The observed allocation and assignment-preference diagnostics ($p = 0.130$ and $p = 0.629$) are compatible with that description. The assignment program was not inspected, so the design interpretation rests on the published account.

The decomposition clarifies the 0.315-point difference between the concordant observational ATE and the unadjusted full-sample randomized estimate. Under the published design, the population composition component is zero; its sample plug-in value is only -0.00016. A further 0.027 points separate the post-stratified and unadjusted randomized estimators. The remaining 0.288-point gap is between the observational ATE and the concordant randomized benchmark, with a percentile interval of $[-0.634, 1.073]$.

The vocabulary analysis adds an outcome-specific comparison. For the same mathematics-preferring target population and corresponding inverse-odds weighting construction, the QED ATU exceeds its randomized benchmark by 0.913 vocabulary-test points, with a nominal 95% interval of [0.115, 1.909]. The mathematics ATT point estimates nearly coincide, whereas this vocabulary ATU contrast is positive. This pattern suggests that the performance of a fixed adjustment strategy can vary across outcomes. It also motivates examining outcome-specific selection mechanisms and outcome-model specification in future work.

Precision depends on the weighting direction. In the mathematics ATT and vocabulary ATU comparisons, 817 participants in G4 provide the weighted source population for a target of mathematics-preferring participants; the effective sample size is about 146. The reverse mathematics ATU comparison has an effective sample size of about 62 among 288 participants and a maximum weight above 50. These diagnostics help explain why the two weighting directions have different precision.

7.1 Limitations

The observational comparisons rely on conditional exchangeability of preference given the measured covariates, positivity, consistency, no interference, and a common treatment version. The randomized design identifies the preference-group benchmarks but does not randomize preference. The vocabulary analysis is post hoc and includes six nominal contrasts across two outcomes and three estimands; its interval is not multiplicity-adjusted. The assignment mechanism is taken from the published study description rather than an inspection of the assignment program. Finally, a second computational implementation reproduced point estimates to numerical tolerance but used a different resampling scheme and random-number stream, so it serves as a numerical cross-check rather than a reconstruction of the original execution.

8 Conclusion

The four-arm design supplies a randomized benchmark for the concordant population at any common assignment probability, and under the published one-half design that benchmark equals the full-population average treatment effect. In the prespecified mathematics comparison, randomized

and observational point estimates are close but the interval for their difference is wide. The finite-sample decomposition attributes little to sample composition and separates the observational ATE–concordant benchmark gap from a small difference between randomized estimators. The post hoc vocabulary analysis identifies a positive ATU discrepancy for the same mathematics-preferring target population under the corresponding weighting construction. Together, the results show why within-study comparisons should align estimands and examine outcome-specific performance rather than treating one close pair of point estimates as a general validation of an observational adjustment strategy.

Generative AI Use Disclosure

Generative AI tools were used to assist with code review, preparation of a second computational implementation, and manuscript language editing. The authors take responsibility for the analyses, interpretations, and final content.

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Supplementary Material
Estimand Alignment and Outcome-Specific Discrepancies in a Four-Arm Within-Study Comparison

Yunjie Guo Hanqing Yang

This supplement provides additional numerical details, estimator formulas, and exploratory displays for the accompanying article. The prespecified primary interval is the one in the article's Table 2. Intervals from the later four-cell bootstrap and from 1,000-replicate trimming runs are identified separately. Tables S1 and S2 report estimates to four decimal places, with the small primary contrast shown to five decimals.

1 Exact primary and full-sample estimates

Table S1 restates the prespecified primary comparison to four decimal places. The contrast interval is the percentile interval from 2,000 successful replicates of the bootstrap that redraws G1, G2, and G4. The analytic standard errors are the two-sample standard error for the randomized difference in means and the cross-fit score standard error for the observational estimator.

Table S1: Prespecified primary estimates, rounded to four decimals.

Quantity	Value
Randomized $\widehat{\tau}_1$	0.7947
Analytic SE	0.2778
Analytic interval	[0.2503, 1.3391]
Observational ATT	0.7942
Score SE	0.2238
Analytic interval	[0.3556, 1.2329]
Difference, observational minus randomized	−0.00045
Prespecified bootstrap SE	0.4217
Prespecified percentile interval	[−0.9492, 0.6638]
Successful replicates	2,000 of 2,000

Table S2 records the full-sample contrast used in the decomposition. Its percentile interval belongs to this contrast, rather than to $\widehat{\tau}_{\mathrm{ATE}}^{\mathrm{QED}} - \widehat{\tau}_C$.

The decomposition uses the concordant benchmark $\widehat{\tau}_C = 0.7872$, the post-stratified benchmark $\widehat{\tau}_{\mathrm{ps}} = 0.7873$, and components 0.2882, −0.0002, and 0.0270. Their sum equals the full-sample discrepancy, 0.3151, up to rounding. Across the 2,000 four-cell replicates, the largest absolute row-wise identity error was below 5×10^{-16}.

The composition interval is [−0.0095, 0.0098]. The weight $\widehat{\pi}_c - \widehat{\pi}$ equals −0.0153 in every replicate. The resulting conditional interval for $\widehat{\tau}_1 - \widehat{\tau}_0$ is approximately [−0.641, 0.622], with point estimate 0.0102. The interval for the post-stratified minus unadjusted randomized term is [0.0238, 0.0301].

Table S2: Full-sample contrast from the earlier secondary analysis, rounded to four decimals.

Quantity	Value
Full-sample randomized difference in means	0.7603
Analytic SE	0.1511
Analytic interval	[0.4642, 1.0565]
Observational ATE	1.0754
Score SE	0.2332
Analytic interval	[0.6183, 1.5325]
Difference	0.3151
Bootstrap SE	0.4578
Percentile interval	[−0.6249, 1.1444]
Successful replicates	2,000 of 2,000

Adjustment covariates

The full 25-variable adjustment set uses the released data-column names: `female`, `white`, `black`, `asian`, `hisp`, `married`, `logAge`, `income`, `collegeS`, `collegeM`, `collegeD`, `calc`, `logBooks`, `mathLike`, `big5O`, `big5C`, `big5E`, `big5A`, `big5N`, `AMAS`, `logBDI`, `MCS`, `GSES`, `vocabPre`, and `mathPre`. The existing log-transformed columns were used as supplied in the released data; they were not reconstructed from the raw age, books, or BDI columns.

2 Vocabulary outcome replication

The post hoc vocabulary analysis reverses the assignment and preference indicators and uses `vocabPost` as the outcome. The concordant sample remains G1∪G4. The QED ATU under the reversed coding targets mathematics-preferring participants (G1), and its randomized counterpart is the vocabulary-minus-mathematics effect among those participants. The four-cell bootstrap resamples G1–G4 at their observed cell sizes for 2,000 replicates.

The vocabulary fit uses the same 25 covariates, five-fold cross-fitting, and fixed seed as the mathematics analysis. For the mathematics-preferring target group, the reversed coding maps the vocabulary ATU odds weight to the mathematics ATT odds weight. Under the same symmetric logistic specification and fold assignment, $\widehat{e}_V(X) = 1 - \widehat{e}_M(X)$, so for G4,

$$\frac{1 - \widehat{e}_V(X)}{\widehat{e}_V(X)} = \frac{\widehat{e}_M(X)}{1 - \widehat{e}_M(X)}.$$

The supplied diagnostics report an effective sample size of 145.9 and a maximum weight of 7.70 for this weighting direction.

Table S3: Post hoc vocabulary outcome estimates and contrasts. Values are rounded to three decimals.

Target group	RCT	QED	Difference	Bootstrap SD	95% percentile interval
Vocabulary preference (ATT)	2.185	2.577	0.392	1.070	[−2.066, 2.197]
Mathematics preference (ATU)	2.039	2.952	0.913	0.463	[0.115, 1.909]
Concordant population (ATE)	2.147	2.674	0.528	0.799	[−1.254, 1.841]

The nominal ATU interval excludes zero; the ATT and ATE intervals include zero. These post hoc contrasts are exploratory and no multiplicity adjustment is applied.

3 Cross-fit scores and the second bootstrap

On an estimation sample of size n, let D_i be the treatment indicator, Y_i the observed outcome, e_i the out-of-fold propensity score, and (μ_{1i}, μ_{0i}) the out-of-fold arm-specific outcome predictions. Clipping replaces e_i by its projection onto $[10^{-6}, 1 - 10^{-6}]$. For the average treatment effect the cross-fit score of Bang and Robins (2005) is

$$(\mu_{1i} - \mu_{0i}) + \frac{D_i}{e_i}(Y_i - \mu_{1i}) - \frac{1 - D_i}{1 - e_i}(Y_i - \mu_{0i}). \tag{S1}$$

For the average treatment effect on the treated, with the sample treated fraction $\widehat{q} = n^{-1} \sum_{i=1}^{n} D_i$ used in place of p,

$$\frac{D_i}{\widehat{q}}(Y_i - \mu_{0i}) - \frac{(1 - D_i)\, e_i}{\widehat{q}\,(1 - e_i)}(Y_i - \mu_{0i}). \tag{S2}$$

The untreated score is the treated score after the treatment labels are reversed. Folds are ordinary K-fold partitions with $K = 5$, not stratified folds. In the prespecified fit the random seed is 20261026. The reported analytic standard error uses the raw scores, with divisor $n - 1$ inside the standard deviation. A recentered influence function that treats $\widehat{q}$ as estimated is a different standard error; it was not the primary reported standard error, and it is not the bootstrap standard deviation.

The second implementation reproduced the primary observational point estimate to within 1.3×10^{-15} and the primary randomized estimate exactly. Its four-cell bootstrap is summarized in Table S4. The contrast standard deviation is about 0.470, and the percentile interval is about $[-0.925, 0.661]$. This bootstrap redraws G1, G2, G3, and G4, whereas the prespecified bootstrap redraws only G1, G2, and G4. The random-number streams also differ, and the observational treated-effect margin has a heavy tail. The interval is an implementation-specific sensitivity result. Table S1 is unchanged by it.

In that margin, 41 replicates of the observational treated estimate are negative and one is below -2. The minimum is about -8.62, and the excess kurtosis is 163.25. The percentile interval is sensitive to this tail. Dividing the interquartile range by 1.349 gives 0.307, which describes the center of the distribution only and is not a replacement interval.

Table S4: Four-cell bootstrap quantities from the second implementation. These values do not replace Table S1.

Quantity	Value
Primary-contrast bootstrap SD	0.4704077152493183
Primary-contrast 2.5th percentile	−0.9251154466640124
Primary-contrast 97.5th percentile	0.6610285346256077
Observational ATT bootstrap SD	0.38874647117306493
Observational ATT 2.5th percentile	0.03410392205623919
Observational ATT 97.5th percentile	1.346552534607815
Minimum observational ATT	−8.621320612422295

Figure S1 plots, for each scalar, the point estimate and the 2.5th and 97.5th percentiles from this same four-cell file. The three pairs are the treated contrast against $\widehat{\tau}_1$, the untreated contrast against $\widehat{\tau}_0$, and the average-effect contrast against $\widehat{\tau}_C$. Each bar is a marginal percentile interval. The figure does not display the prespecified analytic intervals, it does not display the prespecified primary contrast interval, and the three intervals are not a simultaneous band. The axis text generated with the figure describes the intervals as joint-bootstrap percentile intervals. The jointness of the redraw means that the estimates in one row share a sample. The plotted endpoints remain marginal.

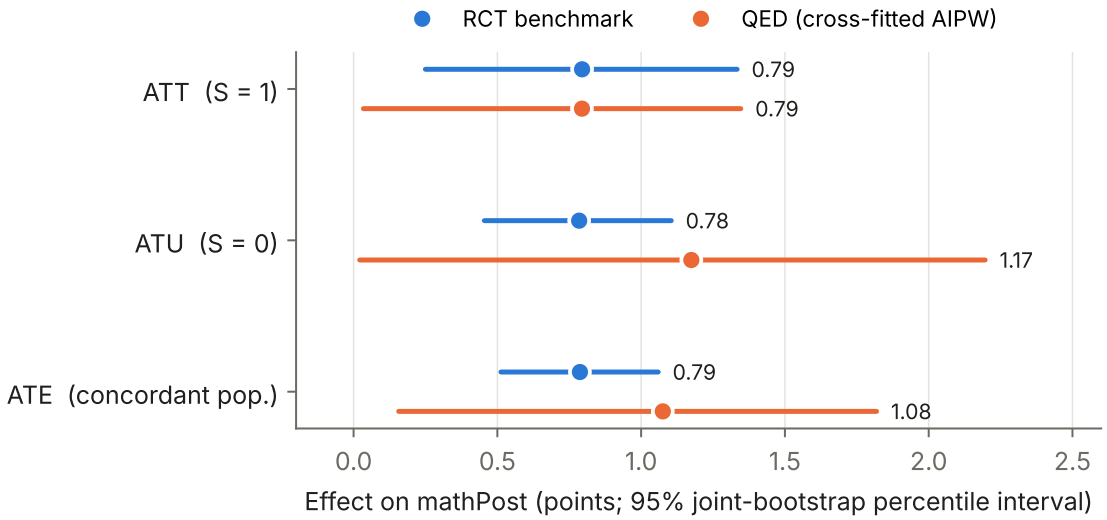


Figure S1: Point estimates and marginal 95% percentile intervals from the four-cell bootstrap. Blue denotes the randomized benchmark and orange the cross-fit observational estimator. The treated observational interval is affected by a replicate near −8.62. These intervals do not replace the prespecified primary interval.

A second computational implementation reproduced the reported point estimates and summaries to numerical tolerance (maximum absolute difference 2.8×10^{-14}). This numerical agreement provides a computational cross-check; the second implementation used a different resampling stream and does not reconstruct the original execution process.

4 Cross-fitting seeds

Figure S2 and Table S5 summarize 200 fold seeds on the fixed sample. The median observational treated estimate is 0.7338163919022771. The prespecified seed's treated estimate is at the 77th percentile of these 200 numbers, and 11.5 percent of the seeds yield an absolute contrast with $\widehat{\tau}_1$ below 0.01. The dashed line in the figure marks that prespecified seed. The solid line marks the randomized benchmark.

Table S5: Observational estimates across 200 fold seeds.

	ATT	ATU	ATE
Median	0.734	1.196	1.076
5th percentile	0.641	1.014	0.942
95th percentile	0.833	1.368	1.195
Minimum	0.580	0.548	0.609
Maximum	0.865	1.453	1.283
Standard deviation	0.061	0.116	0.088

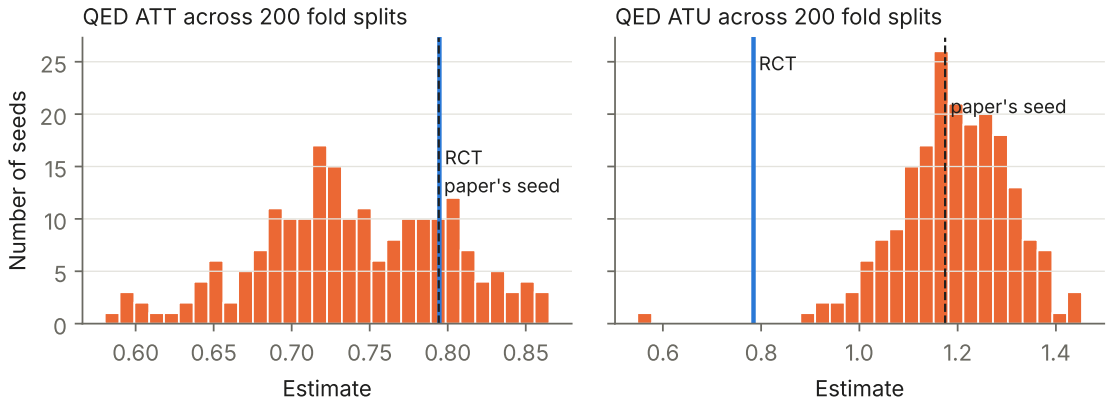


Figure S2: Histograms of the observational treated and untreated estimates over 200 fold seeds. The seeds repartition one sample. They are not replications of the study.

5 Covariate sets

The nested sets in Table S6 and Figure S3 add, in order, eleven demographic and schooling variables, the mathematics and vocabulary pre-tests, four mathematics-motivation measures (`mathLike`, `calc`, `AMAS`, `MCS`), and eight personality and related scales, including `logBooks`. Each nonempty entry is a median over seeds 1 through 50. The empty entry is one concordant difference in means, copied into all three columns. Discrepancies subtract $\widehat{\tau}_1$, $\widehat{\tau}_0$, and $\widehat{\tau}_C$ respectively. The full-set median treated estimate, 0.738, is not the prespecified estimate 0.794.

Table S6: Median observational estimates and discrepancies by covariate set.

Set	ATT	ATU	ATE	ΔATT	ΔATU	ΔATE
None	3.451	3.451	3.451	2.656	2.666	2.663
Demographics	2.952	3.143	3.094	2.157	2.359	2.307
Plus pre-tests	1.594	1.747	1.707	0.799	0.962	0.920
Plus motivation	0.735	1.174	1.066	−0.060	0.389	0.279
Full set	0.738	1.209	1.091	−0.056	0.425	0.304

Table S7 reports the largest median leave-one-out shifts relative to the full-set median. A positive shift means that omitting the named covariate raises the median observational estimate. Omitting `mathPre` shifts the treated median by 0.669 and the untreated median by 0.235. Omitting `mathLike` shifts the treated median by 0.249 and the untreated median by 0.011. These quantities are not an additive decomposition of selection bias.

After weighting, the largest absolute standardized difference in the primary treated comparison is 0.143, on the vocabulary pre-test. Three covariates exceed 0.1 in absolute value, and none exceeds 0.2. In the untreated weighting, the largest absolute standardized difference is 0.292, on agreeableness (`big5A`), and four covariates exceed 0.1. The treated-comparison effective sample size is 145.918 of 817, the maximum control weight is 7.703, and the ten largest control weights account for 19.1 percent of the weight. The untreated-comparison figures are an effective sample size of 61.929 of 288, a maximum weight of 52.572, and a top-ten share of 31.3 percent.

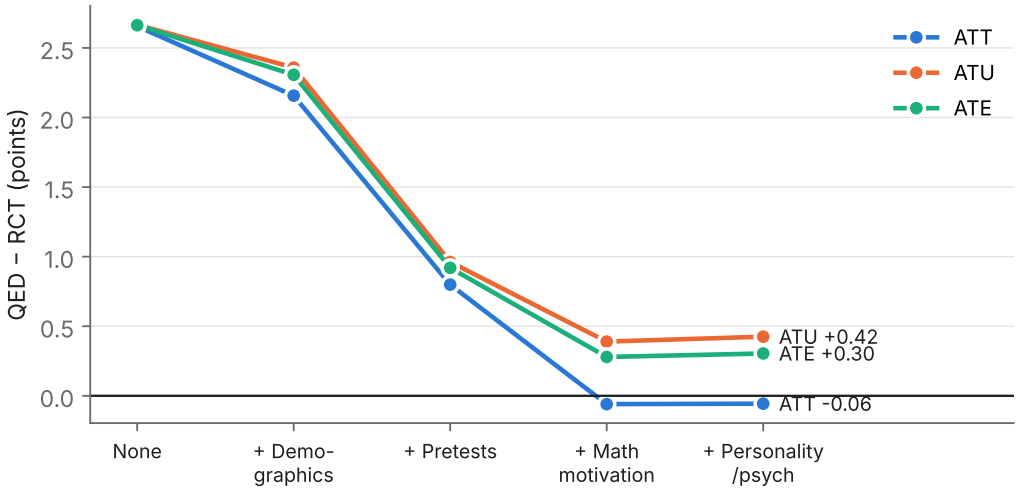


Figure S3: Discrepancies from the randomized benchmarks as covariates are added. The empty-set discrepancies differ across estimands only because the benchmarks differ. The paths are exploratory medians over 50 seeds.

Table S7: Largest median leave-one-out shifts relative to the full covariate set.

Omitted covariate	Treated shift	Untreated shift	ATE shift
`mathPre`	0.669	0.235	0.340
`mathLike`	0.249	0.011	0.072
`MCS`	0.179	0.089	0.107
`AMAS`	0.037	0.189	0.146
`vocabPre`	−0.115	−0.118	−0.117
`big5A`	0.002	−0.101	−0.065

6 Sensitivity grid

Table S8 and Figure S4 report the inverse-probability bounds. The column "RCT inside the point bounds" refers to the numerical bounds at that Λ, not to the percentile interval of the endpoints. At $\Lambda = 1$ the point bounds collapse to the Hájek estimates 0.6857768855171429 and 0.9596311794657346 and do not contain the randomized points. The endpoint intervals at $\Lambda = 1$ do extend below those randomized points. The augmented untreated estimate, 1.1745100278233231, exceeds the inverse-probability value by 0.2148788483575885. The grid value $\Lambda = 1.06$ is not a row of Table S8; it is the first point, on a separate grid with step 0.01, at which the untreated lower point bound is at most $\widehat{\tau}_0$.

7 Trimming

Table S9 reports regions defined by a propensity score fit on the concordant covariates, without the outcome, and then held fixed for the full sample of 2,200. The bootstrap redraws within the region and does not refit the trimming score. Every row is a different target population. The interval in the untrimmed row comes from 1,000 replicates and is not a primary or a 2,000-replicate interval.

Table S8: Inverse-probability sensitivity bounds. Endpoint intervals are percentile intervals of the bound endpoints from 1,000 replicates.

Λ	Estimand	Lower	Upper	Endpoint low	Endpoint high	RCT in bounds
1.00	ATT	0.686	0.686	−0.252	1.287	No
1.00	ATU	0.960	0.960	−0.340	2.007	No
1.10	ATT	0.395	0.983	−0.534	1.590	Yes
1.10	ATU	0.662	1.258	−0.612	2.291	Yes
1.25	ATT	0.016	1.385	−0.887	1.992	Yes
1.25	ATU	0.272	1.657	−0.994	2.637	Yes
1.50	ATT	−0.496	1.968	−1.365	2.565	Yes
1.50	ATU	−0.286	2.203	−1.515	3.140	Yes
1.75	ATT	−0.897	2.455	−1.712	3.027	Yes
1.75	ATU	−0.735	2.676	−1.957	3.601	Yes
2.00	ATT	−1.227	2.876	−1.999	3.435	Yes
2.00	ATU	−1.128	3.061	−2.282	3.936	Yes
2.50	ATT	−1.696	3.542	−2.432	4.088	Yes
2.50	ATU	−1.736	3.697	−2.824	4.462	Yes
3.00	ATT	−2.051	4.051	−2.706	4.578	Yes
3.00	ATU	−2.197	4.154	−3.297	4.854	Yes

The Crump threshold fit on the concordant sample is $\alpha = 0.084311327831958$, reported in the table as 0.084.

Table S9: Exploratory overlap regions. Sample sizes for G1 and G4 are the concordant cells inside the region. Intervals are percentile intervals from the trimming bootstrap.

Region	n	G1 / G4	$\widehat{\tau}_0$	ΔATT	ΔATU	ΔATE
Untrimmed	2,200	288 / 817	0.785	−0.000	0.390	0.288
				[−1.004, 0.682]	[−0.729, 1.427]	[−0.595, 1.088]
[0.05, 0.95]	1,846	281 / 624	0.503	0.120	0.422	0.328
				[−0.891, 0.781]	[−0.483, 1.994]	[−0.349, 1.368]
[0.1, 0.9]	1,446	259 / 447	0.430	−0.231	0.716	0.369
				[−1.276, 0.722]	[−0.918, 2.317]	[−0.697, 1.399]
Crump, $\alpha = 0.084$	1,561	268 / 495	0.404	−0.011	0.640	0.411
				[−1.008, 0.714]	[−1.033, 2.209]	[−0.715, 1.449]

References

Bang, H. and Robins, J. M. (2005). Doubly robust estimation in missing data and causal inference models. *Biometrics*, 61(4):962–973.

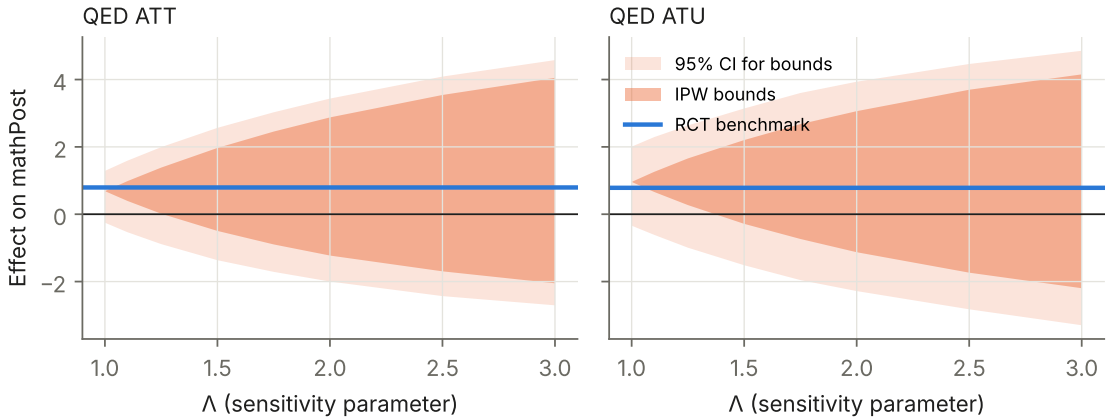


Figure S4: Point bounds and endpoint percentile bands for the inverse-probability functionals, with the randomized benchmark drawn in blue. The darker band is the point-bound interval. The lighter band is not a confidence interval for the causal effect. The bounds are not sharp bounds for inverse-probability weighting, and they do not apply to the augmented estimator.